UV/vis measurements using Avantes spectrometer, Lab 208, CEEC I

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Tags: UV/Vis KRA NB Avantes
Category: Protocol
Created by: Nadzeya Brezhneva

Goal

UV/vis measurements using Avantes spectrometer AvaSpec-ULS-TEC-EVO (light source - AvaLight-DH-S-BAL)

Manuals for the device can be found here: Equipment - Avantes CUV-FL-UV/Vis Cuvette Holder, Equipment - Avantes Fiber Optic UV/Vis System, Equipment - AvaSpec-ULS2048X64TEC-EVO, Equipment - AvaLight-DH-S-BAL.

For questions ask Nadja or Kristína.

Everybody needs an introduction before measuring UV/Vis.

Prerequisites and preparation

- UV/vis spectrometer
- Cuvettes (PS - micro cuvettes (1 ml) or quartz (up to 3.5 ml))
- Analyzed solution, solvent (for reference spectra)
- Eppendorf pipette 100-1000 uL, tips
- Laptop with software Avasoft 8

Steps

Step number	Step description
1	Turn the light source and spectrometer on (O/I button on the back side for both components), switch both lamps (Halogen and deuterated) on the front panel of the light source, wait for ca. 10 min before start of the measurements.
2	Connect USB cabel to the laptop.
3	Start the software (AVAsoft).
4	Prepare a reference cuvette. Place the cuvette with your solvent (reference) into the holder. Turn the measurement to continuous measurement under the "Start" button. Start measurements by pressing the "Start" button. Check the oversaturation value of the lamp spectra in the Scope mode (if necessary, press the button "Scope mode" in the field "Measure mode" the intensity should comprise the value ~ 90%, 59000-60000 counts). If there is oversaturation, reduce the value of integration time; if the intensity is too low, increase integration time.

5	Stop the measurement. Take the cuvette out of the holder. Start the measurement. Press the button "Save dark spectrum" in the field "Simulated device" (the shutter will be closed automatically and the dark spectrum will be saved), stop measurements.
6	Place the reference cuvette into the sample holder.
7	Start new measurements by pressing the "Start" button in the continuous mode, measure the spectrum of the solvent, press the button "Save reference spectrum" to save the spectrum. In order to check the absorbance spectrum, switch to "Absorbance mode" in the field "Measure mode". Stop measurements.
8	Remove the cuvette with the solvent, insert the cuvette filled with the investigated solution. Switch the measurement mode into "Single" (you can find this under the "Start" button). Switch to the Absorbance mode. Start measurement.
9	If there is oversaturation, stop measurements, adjust the concentration of the solution, so that the final spectrum has max. absorbance at ca. 1-1.5 a.u. Important note: besides integration time, it is also possible to adjust spectrum of the analyte with choosing appropriate number of spectra to average (Averaging).
10	Stop measurements. Save the spectrum by pressing the button "Save" in the field "Spectral data" (the spectrum will be saved in the .ABS8 format). Add comments to each spectrum - sample name, conditions of measurements, etc.
11	Convert the obtained .ABS8 files into ASCII in the following way: File -> Convert file to -> ASCII -> Select X-axis units: Wavelength (nm) -> Convert separate files. Select appropriate file(s). The final files will be in .txt format containing comments added to spectrum (at step 10) as headings.
	Important note: If the measurements are planning to be performed within ca. 1 h, there is no need to switch the lamps off. Otherwise (if the interval between the measurements comprises several hours or more), then it makes sense to switch the lamps off.
12	After switching the lamps off, it is necessary to let the ventillation work for ca. 10 min to cool the system. Afterwards, switch the button O/I (on the back panel) off. Turn the spectrometer off.

Plotting

Script for processing data (ASCII files exported from Avasoft software): 241209_Avantes_UV_Vis_data_analysis.py

Python test for plotting a folder with .txt files: plotting_UVVis_folder.py (ask Kristína if you have questions)

If you want to plot two plots next to each other you can use the following script: plotting_UVVis_two_folders.py (ask Kristína if you have questions)

AvaSoft software

Once you download the software you need to change one thing in the settings regarding the O/I shutter.

Go to Settings --> I/O shutter --> put check on both listed under "shutter control"

Tips

- adjust the averaging to 100 before measuring (so you don't forget that)
- always have a cuvette with your reference solution on the side and a second cuvette for the analyte solution (you never know when you need it)
- always wipe your cuvettes with a Kimtech wipe before placing them into the cuvette holder
- if you plan to measure multiple spectra, save them in a folder which you can convert at once afterwards

Linked resources

Equipment - AvaLight-DH-S-BAL

Equipment - Avantes CUV-FL-UV/Vis Cuvette Holder

Equipment - Avantes Fiber Optic UV/Vis System

Equipment - AvaSpec-ULS2048X64TEC-EVO

Equipment - Avantes UV/vis spectrophotometer

Attached files

plotting_UVVis_two_folders.py
sha256: 8034e3c8b6824573f892a8b915e40b1202c9f2303b4758f31f0f07b6eabc3caf

plotting_UVVis_folder.py
sha256: d8ac505e16e2a3d56c3ab1f231543a105f6ce7acb3dd8445ea7159b022263c8c

Unique eLabID: 20250218-362827e1144962fe520a65cfebe5d06163f96b4d
Link: https://elab.water-splitting.org/database.php?mode=view&id=187